

ALI NIKKHAH

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SUMMARY

AI/ML Engineer with 2+ years of experience designing and deploying production agentic systems, multimodal retrieval, and ML infrastructure at scale. At Digikala, led the enterprise AI intelligence layer serving 5M+ daily queries, transitioning from static semantic matching to stateful LangGraph multi-agent pipelines with hybrid RAG (FAISS + Elasticsearch via RRF), HyDE query expansion, and KAG symbolic reasoning. Serving stack: async FastAPI, vLLM (PagedAttention), Triton Inference Server, multi-tier Redis caching. At Advanced Analytics Australia, engineered edge computer vision pipelines on AXIS hardware: TensorRT/TensorFlow Lite compilation, Triton serving, Kafka streaming for 100+ camera feeds. At Turquoise Digital, built BNPL credit risk models (AUC 0.87, -22% default rate), financial agentic chatbots with Whisper/wav2vec2.0 voice transcription, and real-time portfolio allocation with time-series transformers. Strong foundation across the full ML lifecycle: distributed training, MLOps, high-concurrency serving, and ML platform engineering.

TECHNICAL SKILLS

Languages & Runtimes: Python (Asyncio, Pydantic), Go, C++, SQL, Bash, R, TypeScript, Node.js
Deep Learning & Core AI: PyTorch, TensorFlow, JAX, Hugging Face Transformers, Time-Series Transformers, ONNX
Agentic AI & RAG: LangGraph, LangChain, LlamaIndex, CrewAI, AutoGen, HyDE, KAG, Graph RAG, FAISS, ChromaDB, Pinecone
Vision & Audio: OpenCV, ViT, BLIP, T5, Whisper, wav2vec2.0, Librosa, ESPnet, TensorRT, TensorFlow Lite
Data Engineering: Apache Spark (PySpark), Trino, ClickHouse, Airflow, Kafka, PeerDB, Elasticsearch, Parquet, Delta Lake
MLOps & Serving: Ray, Triton Inference Server, vLLM (PagedAttention), TensorFlow Serving, TensorRT-LLM, MLflow, W&B, DVC
DevOps & Infrastructure: Docker, Kubernetes, Terraform, ArgoCD, GitHub Actions, Jenkins, GitLab CI
Databases & Monitoring: PostgreSQL, Redis, Prometheus, Grafana, ELK Stack, Metabase, Great Expectations

PROFESSIONAL EXPERIENCE

Digikala

AI Engineering R&D Lead

Tehran, Iran

Jan 2025 – Aug 2025

- Led the full architectural transition of the enterprise search and knowledge layer from static semantic matching to stateful multi-agent pipelines powered by **LangGraph**; the system now handles over **5 million daily queries** (p99 <200ms) across product discovery, customer support, and internal knowledge retrieval.
- Designed a Modular Control Pipeline (MCP) that cleanly decouples text ingestion, chunking, retrieval, and response synthesis phases, enabling live LLM hot-swapping and zero-downtime vector index migrations without service interruption.
- Built a hybrid retrieval engine combining dense **FAISS** vector search with sparse **Elasticsearch** keyword lookup, fused via Reciprocal Rank Fusion (RRF); implemented **HyDE** query expansion and hierarchical chunking strategies to significantly close the accuracy gap on long-tail and ambiguous product queries (**+23% recall@10**).
- Deployed **Knowledge-Augmented Generation (KAG)** alongside standard RAG to enforce symbolic reasoning constraints on structured product catalog data, substantially reducing factual hallucinations in generated customer-facing answers (**-41% factual error rate**).
- Engineered async **FastAPI** serving layers with multi-tier **Redis** caching and asynchronous request batching; deployed LLMs via **vLLM** with PagedAttention and **Triton Inference Server** to maximize KV-cache utilization and GPU throughput (**3.2x throughput gain, 60% cost reduction per 1K queries**) under burst traffic conditions.
- Supervised Kubernetes deployment of multi-tenant model serving clusters, isolating vision and language model compute nodes for independent horizontal scaling and fault containment (**200 pods, 40% GPU cost reduction**).

Digikala

AI Engineer

Tehran, Iran

Nov 2024 – Jan 2025

- Prototyped and benchmarked RAG architectures against real production query logs; integrated LangGraph cyclic agent graphs for multi-turn session management, dynamic context re-evaluation, and automated query rewriting loops.
- Evaluated retrieval precision and end-to-end latency across candidate retrieval systems; findings directly shaped the R&D roadmap adopted in the subsequent lead role.

Advanced Analytics Australia

Computer Vision Engineer

Remote Contract

Sep 2025 – Apr 2026

- Engineered production video analytics pipelines on AXIS network camera hardware for health, safety, and compliance monitoring across **12 industrial sites**: handwash verification, PPE detection (boots, helmets), face tracking, and restricted-zone entry alerting (**¡50ms inference, 99.2% uptime**).
- Compiled and quantized vision models into **TensorRT** and **TensorFlow Lite** runtimes; served them through **Triton Inference Server** with dynamic batching, substantially reducing CPU overhead on edge nodes by eliminating Python-level inference overhead (**3.5x speedup vs Python**).
- Architected a high-throughput spatial metadata ingestion broker using **Apache Kafka** to buffer concurrent high-frame-rate tracking streams from **100+ cameras (30 FPS each)** without back-pressure on inference workers.
- Implemented dynamic sensor calibration routines that compensate for exposure variance and lighting shifts across indoor and outdoor camera placements, maintaining stable detection thresholds across all deployment sites (**-35% false positives**).
- Built asynchronous Python streaming workers interfacing directly with camera APIs, managing parallel video payload processing without blocking main inference threads.

Turquoise Digital (Firouzeh Digital)

Tehran, Iran

Senior Data Engineer & Data Scientist

Sep 2025 – Mar 2026

- Architected the central enterprise data lake and OLAP warehouse for a high-volume BNPL and stock-trading platform, routing transactional and user-interaction streams from PostgreSQL into **Trino** via **PeerDB** Change Data Capture (CDC) replication for continuous, low-latency synchronization (**¡60s CDC latency, 5TB/day**).
- Designed and maintained complex **Apache Airflow** DAGs orchestrating multi-stage data transformation pipelines, automated cold-data archiving (Parquet/ORC on object storage), and cross-system synchronization workflows across the full data platform.
- Built a real-time statistical inference engine running parallel A/B hypothesis tests on application features, monitoring user-activity telemetry and automatically flagging significance shifts under volatile market conditions (**50+ concurrent experiments, mSPRT sequential testing**).
- Refactored legacy ETL scripts into modular, object-oriented **PySpark** processing pipelines, improving memory management and substantially reducing algorithmic processing complexity over data lake targets (**60% memory reduction**).
- Developed predictive ML models to classify user debt-repayment risk, compute dynamic credit limits, and automate fund allocation or containment decisions for BNPL accounts (**default rate -22%, AUC 0.87**).
- Engineered an enterprise **LangGraph** agentic chatbot handling customer support ticketing, financial literacy instruction, and live portfolio inquiries; integrated real-time **Whisper** and **wav2vec2.0** voice transcription so call-center audio streams feed the agent directly, eliminating manual ticket creation (**-78% ticket volume, 94% transcription accuracy**).
- Built the AI portfolio allocation system: ingested user risk-tolerance questionnaires, fused historical time-series pricing data with streaming news-sentiment embeddings, and trained time-series transformer models to produce portfolio value forecasts with explicit statistical confidence intervals feeding automated trading allocations (**Sharpe +0.42 vs benchmark**).

HomaCloud

Tehran, Iran

MLOps Engineer

Aug 2021 – Feb 2022

- Orchestrated distributed model training pipelines on Kubernetes using Ray for workload scheduling and MLflow for experiment tracking and artifact lineage (**15+ models, 50+ experiments tracked**).
- Built model inference APIs in FastAPI backed by TensorFlow Serving; automated full deployment lifecycle with Docker, ArgoCD, and GitHub Actions, cutting release cycles to under 10 minutes (**99.9% deploy success rate**).

SELECTED PROJECTS & RESEARCH

Enterprise Agentic AI Serving Infrastructure

Digikala

Lead Backend Engineer

2024 – 2025

- Architected the complete serving layer for a multi-agent RAG system at 5M+ daily queries: FastAPI front-end, vLLM/Triton model runtime with PagedAttention, multi-tier Redis cache, and LangGraph session-state management — all deployed on a Kubernetes multi-tenant cluster.

Distributed Intelligent Surveillance Infrastructure

Advanced Analytics Australia

Core Backend & Edge Systems Architect

2025 – 2026

- Designed the full backend architecture for a multi-site industrial safety compliance system: async Python streaming workers feed a Kafka broker, which distributes camera frames to TensorRT/TF Lite inference containers served via Triton, with detection events written to a PostgreSQL audit store.
- Implemented dynamic sensor calibration and continuous SNR monitoring to maintain stable detection thresholds across variable lighting and exposure conditions at outdoor and indoor sites.

AI-Driven BNPL Credit Risk Engine

Turquoise Digital

Senior Data Engineer

2025 – 2026

- Developed and productionized ML models analyzing user debt-repayment histories and behavioral telemetry signals to compute dynamic credit limits; automated Buy Now, Pay Later fund allocation and account containment decisions with full audit logging (**default rate -22%, AUC 0.87**).
- Implemented Parquet and ORC serialization within PySpark processing blocks, optimizing compression ratios and query execution speeds across the central data lake.

Financial Agentic Chatbot with Voice Intelligence

Turquoise Digital

Senior Data Scientist

2025 – 2026

- Engineered LangGraph agentic chatbot with real-time Whisper/wav2vec2.0 transcription, secure function-calling for live equity queries, and compound-profit calculations — without exposing database credentials to LLM context (**-78% manual tickets, 94% transcription accuracy**).

Automated Ultrasound Report Generation

University of British Columbia

ML Researcher

Feb 2024 – Nov 2025

- Developed vision-language architectures combining **ViT**, **BLIP**, and **T5** trained with self-supervised contrastive learning to automatically generate DICOM-compliant radiology reports from raw ultrasound scans.
- Achieved state-of-the-art **BLEU** and **ROUGE-L** scores on the benchmark; preprint in preparation.

Texture-Free Video Motion Classification

L3S Research Center

Research Assistant

Apr 2023 – Feb 2024

- Designed motion classification pipelines using 3D CNNs and transformer encoders over optical flow fields, deliberately suppressing texture cues to force learning of pure human-dynamic representations; achieved SOTA on **UCF-101** and **HMDB-51**.

RESEARCH HIGHLIGHTS

University of British Columbia

Remote Research Collaboration

Research Assistant – Vision-Language Medical Imaging

Feb 2024 – Nov 2025

- Developed vision-language architectures combining **ViT**, **BLIP**, and **T5** under self-supervised contrastive learning to automate generation of DICOM-compliant radiology reports from raw ultrasound images.
- Achieved state-of-the-art **BLEU** and **ROUGE-L** scores on the evaluation benchmark; preprint in preparation.

L3S Research Center (Leibniz University Hannover)

Remote Research Collaboration

Research Assistant – Video Motion Classification

Apr 2023 – Feb 2024

- Designed texture-free motion classification pipelines using 3D CNNs and transformer encoders over dense optical flow fields, deliberately suppressing appearance cues so models learn generalizable human-dynamic representations.
- Achieved state-of-the-art accuracy on **UCF-101** and **HMDB-51** benchmarks.

EDUCATION

Sharif University of Technology

Tehran, Iran

B.Sc. Electrical Engineering – Digital Systems GPA: 3.65 overall, 3.91 major

Sept 2020 – Jul 2024